

NumPy Assignment Questions

1. a = np.array([
 [11,22,33],
 [44,55,66],
 [77,88,99]
])

Extract:

- first column
- last row
- [
 [22 33]
 [55 66]
]

2. a = np.array([
 [10,20,30],
 [40,50,60],
 [70,80,90]
])

Extract:

- first row
- second column
- element 50

3. Create an array from 1 to 12 and reshape it into:

- (3,4)
- (2,6)
- (2,3,2)

4. a = np.array([1,2,3])
b = np.array([4,5,6])

Perform:

- addition
- subtraction
- multiplication
- division
- power of 3
- square

- mean

5. Create a 1D array containing numbers from 1 to 10 using random module.

Create a 2D array of shape (3,3) containing numbers from 1 to 9 using random module.

6. Generate a random array of shape (2,5).

7. Difference between Python list and NumPy array.

8. Convert the following integer array into float datatype:

```
a = np.array([1,2,3,4])
```

9. What is the difference between:

- dtype

- astype()

10. Create a 2D array and print:

- shape

- size

- ndim

- dtype

Change its datatype.

11. Can a 1D array be sliced like:

```
a[:,1]
```

Why or why not?